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Characterisation and cross-amplification of sex-specific genetic markers in Australasian Egerniinae lizards and their implications for understanding the evolution of sex determination and social complexity

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ABSTRACT

Sex is a pervasive factor that underpins functional phenotypic variation across a range of traits. Although sex can usually be distinguished morphologically, in some species this is not possible. The development of genetic markers for sex identification is, thus, key if we are to incorporate sex into an understanding of ecological or evolutionary process. Here we develop genetic markers for the identification of sex within an iconic Australian lizard group, the *Egernia* group, which is notable for its complex social behaviour. We used restriction-site associated DNA sequencing to characterise sex-specific genetic sequences for a key member of the group, *Liopholis whitii*, and designed primers for four of these putative sex-specific sequences. These primers amplified across some, but not all, species of the group. Our results provided several important insights. They suggest conservatism of a XX/XY sex determination system within the group as well as sex-specific genomic regions that appear independent of the conserved genomic regions identified in other skink species. More broadly, the development of sex markers for the *Egernia* group opens up a range of potential research questions related to the role that sex plays in the mediation of social behaviour and, through this, the emergence and stability of social life.

Keywords: *Egernia*, *Liopholis*, RAD-Seq, reptiles, sex chromosomes, sex determination, sex marker, social behaviour.

Introduction

The sex of an individual can have pervasive effects on a range of traits that mediate the ecological and evolutionary trajectory of populations. These effects include growth (Arnold and Griffiths 2003), dispersal (Clobert *et al.* 2001; Trochet *et al.* 2016), as well as social and sexual behaviours (Hamilton 1967; Wilson 1975). In many taxa, sex can be readily inferred from external phenotypic characters. However, when this is not possible, development of widely applicable genetic markers for sex identification becomes crucial (Griffiths *et al.* 1998; Fernando and Melnick 2001; Rovatsos and Kratochvíl 2017; Romanov *et al.* 2019). Such assignment of sex via molecular markers has been instrumental for advancing key areas in evolutionary ecology, including the evolution of sex-determining mechanisms (Ezaz *et al.* 2009*a*; Bachtrog *et al.* 2014), sex-biased dispersal (Prugnolle and de Meeus 2002; Lawson Handley and Perrin 2007), sex allocation (Ellegren and Sheldon 1997; Komdeur and Pen 2002), and for understanding sex-based differences in ecology and behaviour more generally.

Reptiles represent more than 11 000 species across 92 families (Uetz *et al.* 2019) and are widely used as model organisms in ecological and evolutionary studies (e.g. Blackburn 2006; Rheubert *et al.* 2014; Warner *et al.* 2018; Bels and Russel 2019). However, for a number of reptile species, sex can be difficult to establish morphologically, especially before sexual maturation. This is true for both primary and secondary sexual characters.

For example, in some species, female lizards retain hemipenes throughout development and even into adulthood or display erectile structures similar to male hemipenes (hemiclitores; Böhme 1995; Kasperoviczus et al. 2011; Valdecantos and Lobo 2015). To circumvent the challenges associated with identifying sex, sex-specific genetic markers have been developed for a number of reptile groups. However, because of the lability of sex-determining systems, and the subsequent turnover in sex chromosomes, the development of widely applicable genetic markers to sex reptiles has been challenging (Ezaz et al. 2009b; Rovatsos and Kratochvíl 2017), not the least in the highly speciose skinks (but see Kostmann et al. 2021). Therefore, in many reptile systems we lack an understanding of how offspring sex mediates the outcome of a range of ecological and evolutionary processes.

Australian scincid lizards from the Egerniinae subfamily, which comprises more than 60 species (Uetz et al. 2019), hereafter referred to as the '*Egernia* group', represents an outstanding target system for the development of sex markers. First, it is well known for being one of the few reptile groups that exhibit complex social life. This includes a diversity of social complexity from species that are largely solitary to those that form stable social aggregations sometimes involving multiple overlapping generations (Chapple 2003; Whiting and While 2017; While et al. 2019). Sex is considered to play a key role in mediating the outcome of social behaviour because it can influence levels of conflict and cooperation at both the individual and brood level (Frank et al. 1991; Komdeur et al. 1997; Golla et al. 1999; Richardson et al. 2002; Uller 2006; Alonso et al. 2018). However, the role that sex plays in mediating social evolution within the *Egernia* group is poorly understood because female offspring retain their hemipenes until well after birth and thus cannot be distinguished from males (Chapple 2005). Second, the development of sex markers also has the potential to improve our understanding of the sex determining system of this group and thus the phylogenetic distribution and diversity of sex determination systems observed within the Scincidae family (e.g. Kostmann et al. 2021), and squamate reptiles more broadly (e.g. Ezaz et al. 2009a, 2009b; Gamble 2010; Sarre et al. 2011; Gamble et al. 2015). This is particularly pertinent for species such as members of the *Egernia* group that lack heteromorphic sex chromosomes (Donnellan 1985, 1991). Third, the identification of sex at birth provides unique opportunities to not only explore questions related to sex determination but also sex allocation. Reptiles have historically been disregarded in sex-allocation studies (Wapstra et al. 2007) and thus the ability to sex offspring at birth allows the examination of parental investment in sons versus daughters not only in terms of sex ratio but also in terms of investment per sex, which can, in turn, drive the evolution of sex-determining systems (Uller et al. 2007).

The aim of this study was to use restriction-site associated DNA (RAD) sequencing to identify and characterise genetic sex markers for the *Egernia* group and to examine the extent of conservatism of sex-specific genomic regions across the group. To achieve this, we first generated a set of sex markers for one particularly well-studied species of the group, the Whites' skink (*Liopholis whitii*). We then tested whether these markers cross-amplified across 13 other species within the *Egernia* group), spanning five of the six genera. We also tested the cross-amplification of these markers outside the *Egernia* group, in three skink species within the Scincidae family but from two other subfamilies, Mabuyinae and Eugongylinae.

Materials and methods

Development and validation of markers for *Liopholis whitii*

Voucher specimen and tissue sample collection for initial marker development

Voucher specimens of 31 adult individuals (15 males and 16 females) of *L. whitii* were collected from populations on the eastern coast of Tasmania, Australia (−42.5593352, 147.8487648). On capture, individuals were sexed morphologically via eversion of their hemipenes, a method that is reliable for adults but not offspring (Chapple 2005). Because these samples were taken from males and females from a well studied, long-term, population of *L. whitii* (see below) in which females are monitored for birth and males for paternity of offspring, genetic sex could be further verified. Each individual had a tissue sample taken from the distal end of their tail for DNA extraction. DNA was extracted using DNeasy blood and tissue kit (Qiagen, USA). These samples were used for discovery of single-nucleotide polymorphism via RADseq (see below).

RAD sequencing

We implemented double-digest restriction site-associated DNA sequencing (ddRAD-Seq). DNA samples were digested with two restriction enzymes, namely, *Eco*RI High-Fidelity and *Msp*I (New England Biolabs, USA), and then followed a standard protocol (Peterson et al. 2012). We took 500 ng of DNA for each individual with concentration >20 ng μL^{-1} (quantified using Qubit fluorometer; Life Technologies, USA). Fragment size was selected between 300 and 700 bp and amplified with Q5 high-fidelity DNA polymerase (New England Biolabs, USA) for 12 cycles by using universal sequencing primers. The prepared libraries were sequenced paired-end with length of 150 bp on an Illumina HiSeq 2500 platform in Novogene Company Limited (Hong Kong). Sequence data were deposited in NCBI Short Reads Archive (SRA), with Accession number PRJNA722610.

Primer development and preliminary testing

We used STACKS (v. 1.47, Catchen *et al.* 2011) to process the RAD-Seq reads for each individual. STACKS provides a modular pipeline to work with restriction enzyme-based data. The first module 'process_radtag' was used to filter sequence reads with low quality (Phred score < 30), ambiguous base calls, and incomplete barcode or restriction sites. Next, we used 'ustacks' and 'cstacks' to *de novo* assemble the reads and build a set of consensus loci for the samples together. We then identified those consensus loci that mapped to the sequence reads of all 15 males but none of the 16 females. Finally, we designed polymerase chain reaction (PCR) primers for fragments from those loci by using Primer3web (v. 4.1.0; Koressaar and Remm 2007). Sequence reads were uploaded into Blastn via the basic local alignment search tool (BLAST, https://blast.ncbi.nlm.nih.gov/Blast.cgi) and checked against the NCBI nucleotide and whole-genome shotgun contig databases to identify similarities with existing genome sequences. We also checked the sequence reads against the *Anolis carolinensis* (Alföldi *et al.* 2011) and *Podarcis muralis* (Andrade *et al.* 2019) genomes available on ensemble.

In total, four primer pairs were developed for *L. whitii* by using the initial 31 samples (Table 1). Preliminary tests were conducted to establish whether the primers amplified and to determine the best conditions for amplification. For PCR amplification, each 25 μL reaction contained 1 μL of DNA solution, 12.5 μL of *Taq* 2× Master Mix, and 11.5 μL of ddH_2O. Primer pairs were developed under the following PCR conditions. Cycling parameters for Primer pair 2 are as follows: Step 1 at 95°C for 5 min, Step 2 at 95°C for 30 s, Step 3 at 59°C for 30 s, Step 4 at 72°C for 1 min, Steps 2–4 repeated for 38 cycles, Step 5 at 72°C for 10 min and Step 6, where they were held at 4°C until removed for further processing. Cycling parameters for Primer pairs 12, 14 and 18 are as follows: Step 1 at 98°C for 5 min, Step 2 at 98°C for 30 s, Step 3 at a 60.5°C for 30 s, Step 4 at 72°C for 1 min, Steps 2–4 repeated for 35 cycles, Step 5 at 72°C for 10 min and Step 6, where they were held at 4°C until removed for further processing. The resulting amplicons were then visualised with UV light after migrating for 45 min at 80 V on 2% agarose gels stained with 5 μL Gel Red DNA stain (see examples for each primer pair in Supplementary Fig. S1).

Table 1. Characterisation of four primer pairs for identification of sex in *Liopholis whitii*.

Primer ID	Primer sequence 5′–3′	Tm (°C)	Ta (°C)
2-F	AGC ACA GGA CAC TGC TGT TG	58.0	59.0
2-R	TCC TGG TCA TCC AGG GTA TC	55.7	59.0
12-F	TGA GTG GTT TAT GTG TCT GAC TCT	55.5	60.5
12-R	CTG GTG AAG TCC CTC AGA GC	57.3	60.5
14-F	TGG CCA GCA CTG TAG ATG TC	57.1	60.5
14-R	CAA ATT TTC CCA CTG CCA CT	53.8	60.5
18-F	AGG CGG ATG ACA AAT AGA CG	54.8	60.5
18-R	GCA ATT TGA ATG GGC TTC AT	51.9	60.5

Note: F stands for 'forward' and R for 'reverse'. Primers 2-F and 2-R constitute one primer pair. 'Tm' and 'Ta' are the melting and annealing temperatures respectively.

Field validation of *L. whitii* markers

To further validate the application of our markers, we tested them on individuals from a long-term study population of *L. whitii* (e.g. While *et al.* 2011, 2014). This population is monitored each year, with all adults in the population being morphologically sexed on capture via eversion of the hemipenes. At the end of gestation each year, females are brought into the laboratory to give birth. At birth, offspring tail tips are collected for DNA extraction. Offspring are then released back into the natural population and monitored throughout their life. To further confirm whether our genetic markers successfully identified the sex of individuals, we targeted 20 adult individuals (10 males and 10 females total) from this long-term population who had had DNA extracted as offspring (when sex could not be determined morphologically) and who had survived until adulthood and had thus been subsequently sexed morphologically. For PCR amplification, each 12 μL reaction contained 1 μL of extracted DNA, 5 μL of MyTaqHS, 5.5 μL of MilliQ-H_2O, 0.25 μL of forward primer and 0.25 μL of reverse primer. PCR conditions were the same as those outlined in 'Primer development and preliminary testing'.

Cross-amplification of the markers

We tested whether the four sex-specific primers developed for *L. whitii* cross-amplified in 13 other members of the *Egernia* group belonging to five genera (Gardner *et al.* 2008; see Table 2 for details on the species and their sample details) and in more distantly related Australian skinks (n = 3 species from two genera spread across two subfamilies).

Tissue samples for 13 of the additional 16 species were acquired from collections at the South Australian Museum. These individuals were sexed by visual inspection of ovaries or testes. Corresponding frozen liver-tissue samples were then taken from the Australian Biological Tissue Collection (ABTC) and DNA was extracted using the Gentra method (Gentra® Puregene®). Tissue samples for *Egernia striolata* were taken from individuals caught in the field near Albury in New South Wales (−36.0089581, 146.9713438). DNA was extracted from tail-tip samples using GenCatch™ Blood and Tissue Genomic Mini Prep Kits (Epoch Life Science, Inc., Sugarland, Texas, USA). Tissue samples for the two *Carinascincus* species were also taken from individuals caught in the field in Tasmania. Samples

Table 2. Cross-amplification of primer pairs across *Egernia* and non-*Egernia* skinks.

Species	Sample size		Primer pair 2 Ta (°C)	Primer pair 12 Ta (°C)	Primer pair 14 Ta (°C)	Primer pair 18 Ta (°C)
	Males	Females				
Egernia depressa	2	1	59.0	60.5	–	–
Egernia kingii	2	2	59.0	60.5[A]	NA	–
Egernia richardi	2	2	–	–	–	–
Egernia cunninghami	2	2	59.0	60.5	–	60.5
Egernia striolata	2	2	–	58.3[A]	–	60.5
Egernia stokesii	2	2	59.0	59.0	–	60.5
Cyclodomorphus venustus	2	2	59.0	60.5	–	60.5
Cyclodomorphus melanops	2	2	59.0	60.5	–	60.5
Tiliqua rugosa	2	2	59.0	60.5	–	–
Tiliqua adelaidensis	2	2	59.0	59.0	–	60.5[A]
Tiliqua scincoides	2	2	59.0	59.0[A]	–	60.5[A]
Liopholis inornata	2	2	–	60.5	–	59.0
Liopholis whitii	25	26	59.0	60.5	60.5	60.5
Lissolepis coventryi	2	1	–	–	59.0[A]	–
Eutropis multifasciata	2	2	–	–	–	–
Carinascincus ocellatus	2	2	–	–	–	–
Carinascincus greenii	2	2	–	–	–	–

Note: temperatures represent the annealing temperature at which amplification was successful.
[A]Species for which there was a putative band for one or both females, but the primer pair was able to distinguish between the sexes. Further details with respect to the nature of these differences are provided in Supplementary Fig. S2.

for *Carinascincus ocellatus* were taken from populations in Miena (−41.9813614, 146.7305418) and samples for *Carinascincus greeni* were taken from populations in Ben Lomond (−41.5333266, 147.6651172). DNA was extracted from tail-tip samples by using the same extraction method as outlined for *L. whitii* (see above).

We tested each of the primer pairs developed for *L. whitii* on individuals of known sex from each of the 16 species. PCR plates were run in a thermal cycler set initially with the *L. whitii* cycling parameters as provided above. PCR products were finally visualised with UV light after migrating for 60 min at 90 V on 1% agarose gels stained with Midori Green Advance DNA Stain. Where these initial conditions failed to generate clear results, we ran gradient PCRs to determine the best annealing temperatures required for differentiation between the sexes for each species. Table 2 displays the annealing temperatures for which amplification was the most successful on each primer pair for each species.

Results

Development and validation of markers for *Liopholis whitii*

In total, we obtained 151 811 consensus loci by RAD-Seq for *Liopholis whitii*, of which 98 loci were specific to the 15 males. This excess of male-specific RAD markers is consistent with male heterogamety and therefore with a XX/XY system of sex determination. We found one significant similarity between our sequence reads and an available genome sequence. Specifically, the similarity was found with contig 154 517 (bit score = 93.5; *e*-value = 3×10^{-17}) of the *Gekko japonicus* draft genome (Liu *et al.* 2015). We designed primers for four of the putative sex-specific consensus sequences (2, 12, 14 and 18; Table S1). These primer pairs were then validated on 20 *L. whitii* individuals of known sex (10 males and 10 females). All four primer pairs successfully identified offspring sex across all samples, producing a band on the agarose gel for males and no band for females.

Cross-amplification of the markers

We tested the four primer pairs on 16 skink species, including 13 from the *Egernia* group and three outgroup species. The success of cross-amplification depended both on the primer pair and species (Table 2). Primer pair 12 successfully identified sex for 11 of the remaining 13 species of the *Egernia* group (all species except *Egernia richardi* and *L. coventryi*). Primer pair 18 successfully identified sex in eight of the remaining 13 species of the *Egernia* group. Primer pair 2 successfully distinguished sex in nine of the remaining 13 species within the *Egernia* group. In most species, females exhibited non-target bands of a different weight different (>1000 bp) from that of males; thus, sex could still be distinguished (see Supplementary Fig. S2 for full details). Primer pair 14 successfully identified sex in one additional *Egernia* group species, *L. coventryi*, although the female displayed a putative band of weight similar to that of males. We found no evidence that any of our primer pairs cross-amplified in the three skink outgroup species (*Eu. multifasciata*, *Ca. ocellatus* and *Ca. greenii*).

Discussion

We developed four primer pairs to identify sex in the Australian skink, *Liopholis whitii*, an emerging model system for understanding the origins and evolution of sociality (Whiting and While 2017; While *et al.* 2019). We found that some, but not all, of the primer pairs also cross-amplified in other *Egernia* group species, with one pair cross-amplifying across almost all but the most distantly related members (e.g. *Lissolepis coventryi*). These results suggest that some sex-specific genomic regions are well conserved within the *Egernia* group. All primer pairs failed to amplify across a broader set of skink species outside the Egerniinae subfamily, suggesting that these markers have limited applicability as a skink (family Scincidae) sex marker.

The identification of sex-specific markers for members of the *Egernia* group has the potential to provide several crucial insights into their ecology and evolution. First, they provide insights into the sex-determining system of *Egernia* group skinks. The excess of male-specific RAD markers in *L. whitii* is consistent with male heterogamety and, therefore, with a XX/XY system of sex determination (see also Cooper *et al.* 1997; Stow *et al.* 2001). Cooper *et al.* (1997) found evidence that males are the heterogametic sex in *T. rugosa* and the sex-linked microsatellite locus that supported that finding was subsequently used to complement the anatomical sexual identification of *E. cunninghami* (Stow *et al.* 2001), which further suggests that males are the heterogametic sex in this group.

The amplification of our sex markers across species suggests not only that this XX/XY system of sex determination is conserved across the group but also that specific sex-specific genomic regions themselves are partially conserved. Such conservatism in sex-linked regions has been suggested to be common within skinks. Indeed, Kostmann *et al.* (2021) recently showed significant conservatism in X-linked genes across 13 different skink species spanning 85 million years of evolution. Further work has discovered shared Y-linked sequences between *Eulamprus heatwolei* and *Carinascinsus ocellatus* whose origin is estimated to be between 79 and 116 million years ago (Cornejo-Páramo *et al.* 2020). Given the *Egernia* group sits between *Eu. heatwolei* and *Ca. ocellatus* phylogenetically (Pyron *et al.* 2013), the identification of similar regions in the *Egernia* group would suggest deep conservatism in Y-linked chromosomal regions across the Scincidae. However, our sex markers failed to successfully identify sex in *Carinascincus* (and *Eutropis*) species. This is consistent with recent work by Hill *et al.* (2021), which failed to detect homology of the sex-linked *Ca. ocellatus* region with *L. whitii*. Furthermore, Kostmann *et al.* (2021) found that genes that exhibit strong X-linked conservatism across skinks tended to be more consistent with autosomal, as opposed to sex, chromosomes within members of the *Egernia* group. These differences could be explained by translocations of genes from an X-specific position to an autosomal position or by fusions of autosomes and sex chromosomes (e.g. Sigeman *et al.* 2020; Kostmann *et al.* 2021). A key feature of the *Egernia* group radiation has been the addition of an acrocentric chromosome, resulting in a diploid chromosome compliment of 32 as opposed to 30 in *Eu. heatwolei* and *Ca. ocellatus* (Donnellan 1991; Hill *et al.* 2021). Such karyotype evolution may have mediated the evolution of sex chromosomes within the *Egernia* group as has been suggested for other species (Srikulnath *et al.* 2019; Alam *et al.* 2020). Clearly, more work is required to identify the mechanisms of sex chromosome divergence between the *Egernia* group and other scincid species. However, these initial results suggest that the *Egernia* group may provide an intriguing model system for understanding sex chromosome evolution.

The development of sex-specific markers also has the potential to provide downstream benefits to a range of other research areas. Before applying these markers to specific research questions, it is important to acknowledge that the nature of the sex markers developed, exhibiting a band for males and no band for females, has the potential to produce false negatives as a result of a PCR failure. Thus, utilisation of these markers for downstream study should include appropriate controls. Sex markers have been used to gain a deeper understanding on sex allocation and sex ratio adjustment (Ellegren and Sheldon 1997; Komdeur and Pen 2002; Wapstra *et al.* 2007) but also in conservation programs to effectively manage population sex ratios and enable recovery of endangered species (Griffiths and Tiwari 1995; Miyaki *et al.* 1998; Literman *et al.* 2017). In the context of the *Egernia* group, the development of sex markers is particularly useful to researchers studying the ecology and evolution of social behaviour, for which the *Egernia* group has provided important insights (e.g. Gardner *et al.* 2016; Halliwell *et al.* 2017; While *et al.* 2019). Several species of the *Egernia* group are also of interest from a perspective of reproductive biology, because they are among the very few viviparous vertebrates to give birth to young asynchronously (over up to 10 days) (Chapple 2003; While *et al.* 2007). This has the potential to co-vary with patterns of sex allocation because it has been shown in some birds that females alter hatching sequence in response to offspring sex, thus mediating the effects of hatching asynchrony on offspring fitness (e.g. Badyaev *et al.* 2002; Arnold and Griffiths 2003; Bonisoli-Alquati *et al.* 2011). Whether similar sex-specific patterns of birth occur in the *Egernia* group is unknown but has the potential to provide a fruitful avenue for future research. Finally, the *Egernia* group contains a number of endangered species, including the Alpine she-oak skink (*Cyclodomorphus praealtus*), the guthega skink (*Liopholis guthega*), the pygmy bluetongue lizard (*Tiliqua adelaidensis*), the Arnhem Land gorges skink (*Bellatorias obiri*), and the eastern mourning skink (*Lissolepis coventryi*), for which the ability to sex individuals in a non-intrusive way would provide important conservation tools for monitoring population sex ratios or for population management.

Supplementary material

Supplementary material is available online.

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Ethical approval. All research associated with this paper abides by the *Australian code for the care and use of animals for scientific purposes* (2013). All procedures performed involving animals were in accordance with the ethical standards of the University of Tasmania, Australia.

Data availability. The data that support this study are available in the article and accompanying online supplementary material.

Conflicts of interest. The authors declare no conflicts of interest.

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Supplementary Material

Characterisation and cross-amplification of sex-specific genetic markers in Australasian Egerniinae lizards and their implications for understanding the evolution of sex determination and social complexity

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Table S1. Sequence information for the four Y-linked consensus loci

Y-linked consensus loci	Sequences
2	AATTCAGCACAGGACACTGCTGTTGTGAATAAAGTGGGCATCAGGGGAACAAATGAGTTACTATTTTGT GGACCAAGGTCCAGGGTGATCCACAGGCTTACAGAAACAGAGAGAAAATGGGAAGGTGGCAGGAGATG GGAGGGAGNNNNNNNNNNACTTGTGATGTTAGGGAGATGATACCCTGGATGACCAGGATAACTAATGG AAAATGTTGGTTTCCATAATCTCGTAGCAGAACTGACACCCTGTTACCAACTGTCATCCTGAACCATCTTC AGCTGCAGGGTGGTGCCAGATTTATACCG
12	AATTCTTCTAAACTAAAATTCCTATAGGTAAGAACAGTGGAATATACTGAGTGGTTTATGTGTCTGACTC TATTATCTCTACAGTTTCTGGCTGCATGTAACATCCATTCCATATGTCTTTTAAAATACTGAGAATTGTGT TATCTCTTCATCTCCCCCTGCACTATAGTGCTGAAGAGTGTAAACAAGTTTGCTGGGCTCTGAGGGACTTC ACCAGGTTGTTCCG
14	AATTCATACTGGCCAGCACTGTAGATGTCCCTCAGATGCTATTCTTTCCCATGCTCAGCCCTTTAATTATC CTTTTCAGCCTTGTGGACAAATCGCTCTCTGTAGGTTCCTTTATTCATAATTTCTGAGTTCTAAGGGCCAT CTCAGAGTGGCAGTGGGAAAATTTGTTGGACTGGGTAAGAGGACTCTGGAGAATTATCTATATTCTCCAG AGAGCCG
18	AATTCGGAACGGAGGGGACAGCGCTACAGGAGGTAGCCATGGCCAGATGGTAGTGGACGAAGCGGCAA CAAAGGCGGATGACAAATAGACGACAGCCGAGTGGAAAGTATTTATTTCTACGCACGCGCGATACTCGC ATAAGCTCCACCCCCTGTGCATCAGAGACTGAATGCAGGAGCGCCTCTCAAAATCGCGGTGCTGGCAGA ACATGAAGCCCATTCAAATTGCTCCGTTTGAGGCAGCGGAGGTGGGACTTCCG

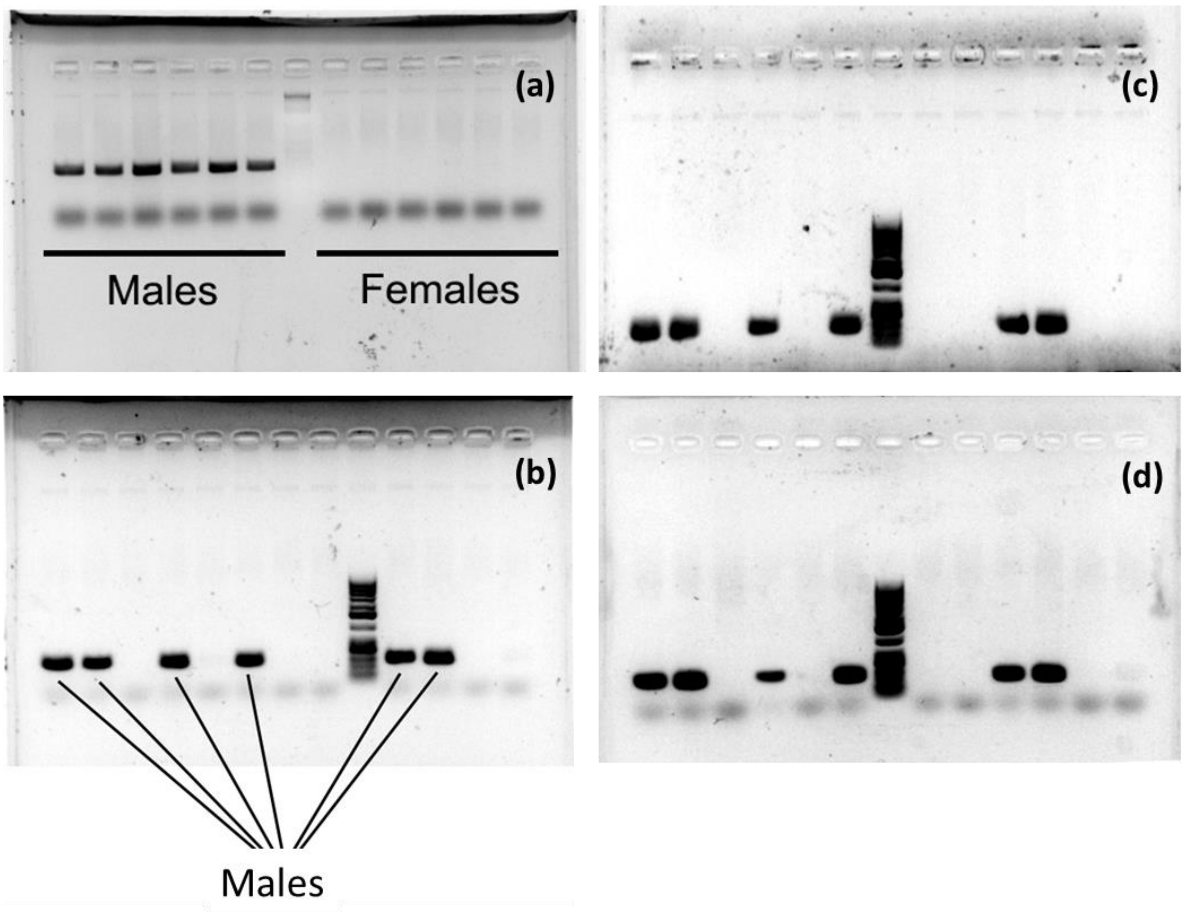
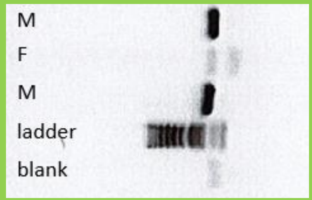
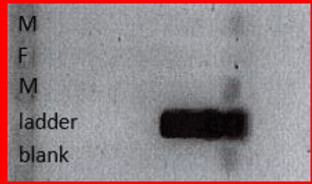
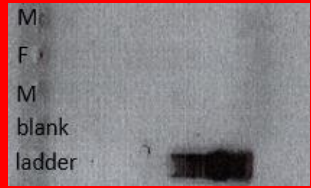
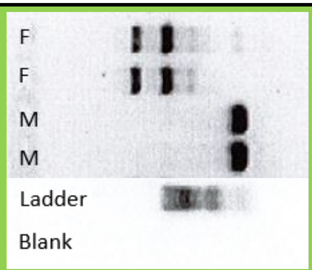
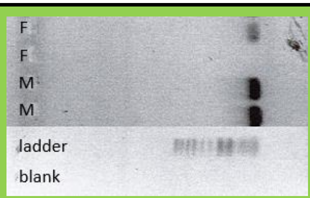
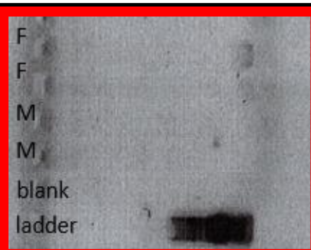


Figure S1: Gel images for the tests of the four primer pairs on 6 male and 6 female *Liopholis whitii*. Figures refer to the gel results from primer pairs 2 (a), 12 (b), 14 (c) and 18 (d). Sexes are grouped together on the gel set for primer set 2 but distributed across the gel for the other primer pairs (male location the same for each set as indicated in Figure 1b).

<table>
<tr><th></th><th colspan="2">Primer Pair 2</th><th colspan="2">Primer Pair 12</th><th colspan="2">Primer Pair 14</th><th colspan="2">Primer Pair 18</th></tr>
<tr><td rowspan="4">Egernia depressa</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><td>Ta tested (°C)</td><td>Ta amplified (°C)</td><td>Ta tested (°C)</td><td>Ta amplified (°C)</td><td>Ta tested (°C)</td><td>Ta amplified (°C)</td><td>Ta tested (°C)</td><td>Ta amplified (°C)</td></tr>
<tr><td>56.8 - 59</td><td>59</td><td>60.5</td><td>60.5</td><td>55 – 60.5</td><td>NA</td><td>55 – 60.5</td><td>NA</td></tr>
<tr><td colspan="2">Notes: Bands for males, no bands for females</td><td colspan="2">Notes: Bands for males, no bands for females</td><td colspan="2">Notes: No bands</td><td colspan="2">Notes: No bands</td></tr>
<tr><td rowspan="4">Egernia kingii</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr>
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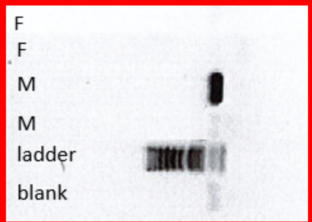
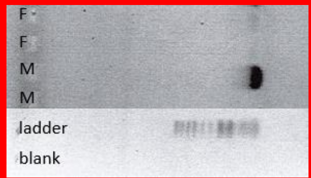
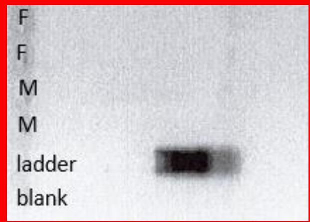
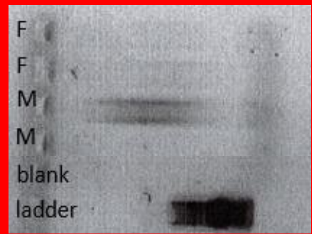
<tr><td>55.4 - 59</td><td>55.4 - 59</td><td>60.5</td><td>60.5</td><td>NA</td><td>NA</td><td>55 – 60.5</td><td>NA</td></tr>
<tr><td colspan="2">Notes: Bands for males, non-target band (<1000 bp) for female</td><td colspan="2">Notes: Bands for males, putative band for one female</td><td colspan="2">NA</td><td colspan="2">Notes: No bands</td></tr>
<tr><td rowspan="4">Egernia richardi</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr>
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<tr><td>56.8 - 59</td><td>NA</td><td>59 – 60.5</td><td>NA</td><td>55 – 60.5</td><td>NA</td><td>55 – 60.5</td><td>NA</td></tr>
<tr><td colspan="2">Notes: Band for one male, no band for the other male or females.</td><td colspan="2">Notes: Band for one male, no band for the other male or females.</td><td colspan="2">Notes: No bands</td><td colspan="2">Notes: No bands</td></tr>
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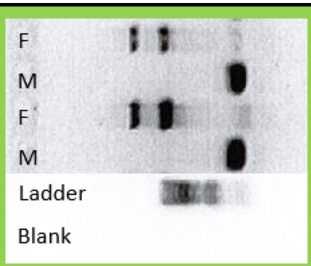
<table>
<tr><th></th><th colspan="2">Primer Pair 2</th><th colspan="2">Primer Pair 12</th><th colspan="2">Primer Pair 14</th><th colspan="2">Primer Pair 18</th></tr>
<tr><td rowspan="4">Egernia cunninghami</td><td colspan="2">
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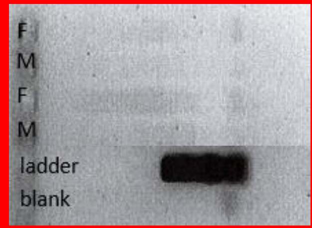
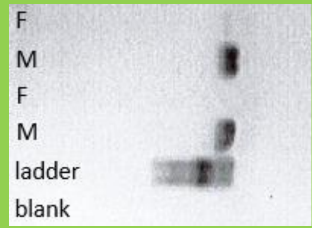
</td><td colspan="2">
</td><td colspan="2">
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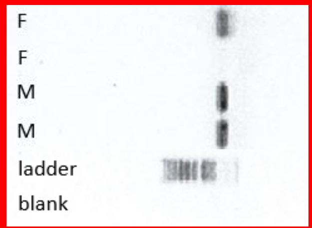
<tr><td>55 – 60.5</td><td>55 - 59</td><td>60.5</td><td>60.5</td><td>55 – 60.5</td><td>NA</td><td>58.5 – 60.5</td><td>60.5</td></tr>
<tr><td colspan="2">Notes: Bands for males, non-target band (<1000 bp) for female</td><td colspan="2">Notes: Bands for males, no bands for females</td><td colspan="2">Notes: No bands</td><td colspan="2">Notes: Bands for males, no bands for females</td></tr>
<tr><td rowspan="4">Egernia striolata</td><td colspan="2">
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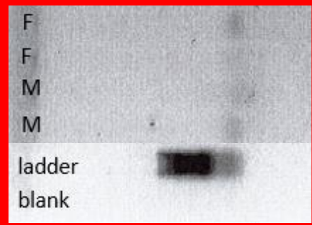
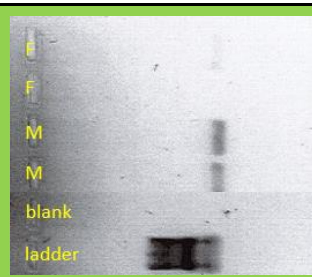
</td><td colspan="2">
</td><td colspan="2">
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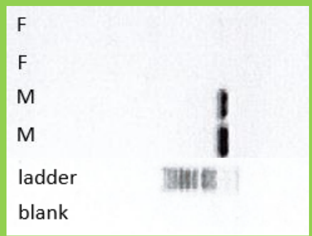
<tr><td rowspan="4">Egernia stokesii</td><td colspan="2">
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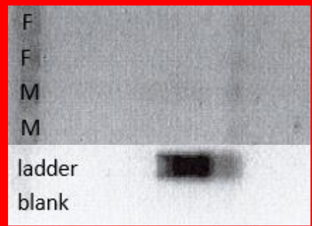
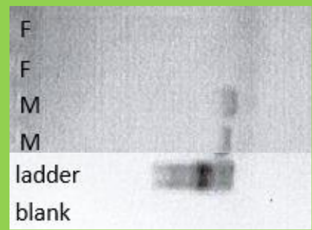
</td><td colspan="2">
</td><td colspan="2">
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<tr><td>59</td><td>59</td><td>59 – 60.5</td><td>59</td><td>55 – 60.5</td><td>NA</td><td>58.5 – 60.5</td><td>60.5</td></tr>
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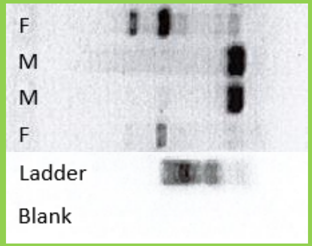
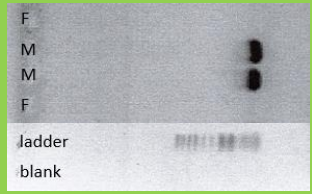
Cyclodomorphus venustus

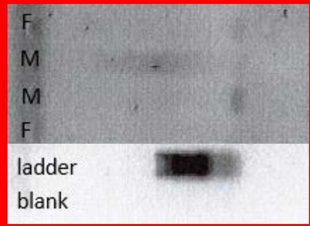
Primer Pair 2



Primer Pair 12



Primer Pair 14



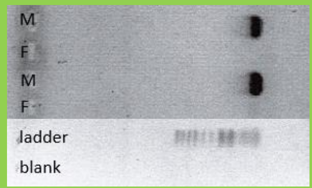
Primer Pair 18

	Primer Pair 2	Primer Pair 12	Primer Pair 14	Primer Pair 18
Ta tested (°C)	58 – 59	60.5	55 – 60.5	58.5 – 60.5
Ta amplified (°C)	59	60.5	NA	58.5 – 60.5
Notes	**Notes**: Bands for males, non-target band (<1000 bp) for female	**Notes**: Bands for males, no bands for females	**Notes**: No bands	**Notes**: Bands for males, no bands for females

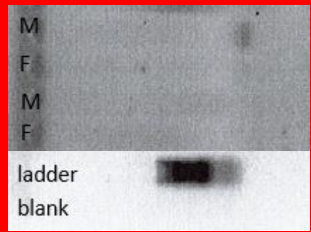
Cyclodomorphus melanops

Primer Pair 2

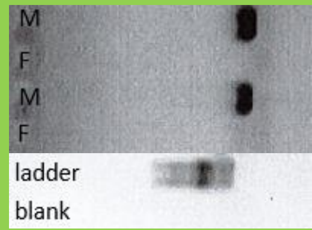
Primer Pair 12



Primer Pair 14



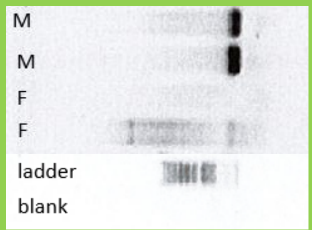
Primer Pair 18



	Primer Pair 2	Primer Pair 12	Primer Pair 14	Primer Pair 18
Ta tested (°C)	59	60.5	55 – 60.5	58.5 – 60.5
Ta amplified (°C)	59	60.5	NA	58.5 – 60.5
Notes	**Notes**: Bands for males, non-target bands (<1000bp) for female	**Notes**: Bands for males, no bands for females	**Notes**: No bands	**Notes**: Bands for males, no bands for females

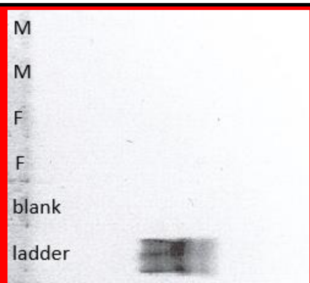
Tiliqua rugosa

Primer Pair 2

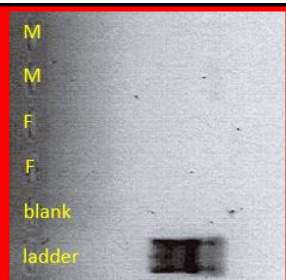


Primer Pair 12

Primer Pair 14



Primer Pair 18



	Primer Pair 2	Primer Pair 12	Primer Pair 14	Primer Pair 18
Ta tested (°C)	58 – 59	58.3 – 60.5	55 – 60.5	58.3 – 60.5
Ta amplified (°C)	59	58.3 – 60.5	NA	NA
Notes	**Notes**: Bands for males, no bands for females	**Notes**: Bands for males, no bands for females	**Notes**: No bands	**Notes**: No bands

<table>
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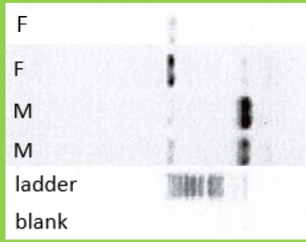
<tr><td rowspan="4">Tiliqua adelaidensis</td><td colspan="2">
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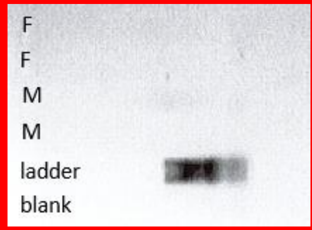
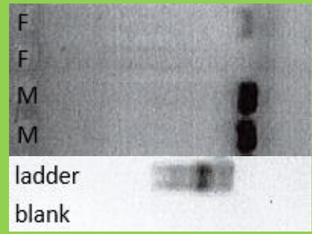
</td><td colspan="2">
</td><td colspan="2">
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<tr><td>55.4 - 59</td><td>55.4 - 59</td><td>59 – 60.5</td><td>59</td><td>55 – 60.5</td><td>NA</td><td>58.5 – 60.5</td><td>58.5 – 60.5</td></tr>
<tr><td colspan="2">Notes: Bands for males, non-target band (<1000 bp) for female</td><td colspan="2">Notes: Bands for males, no bands for females</td><td colspan="2">Notes: No bands</td><td colspan="2">Notes: Bands for males, putative band for one female</td></tr>
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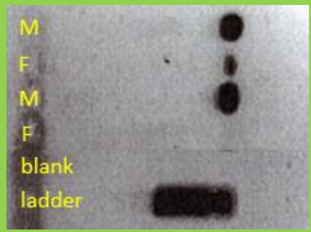
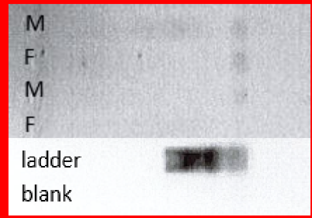
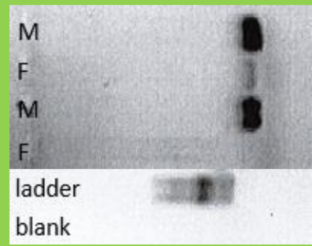
</td><td colspan="2">
</td><td colspan="2">
</td><td colspan="2">
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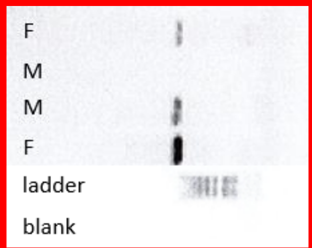
<tr><td>55 – 60.5</td><td>59</td><td>58 – 60.5</td><td>59</td><td>55 – 60.5</td><td>NA</td><td>58.5 – 60.5</td><td>58.5 – 60.5</td></tr>
<tr><td colspan="2">Notes: Bands for males, non-target band (<1000 bp) for female</td><td colspan="2">Notes: Bands for males, putative band for one female</td><td colspan="2">Notes: No bands</td><td colspan="2">Notes: Bands for males, putative band for one female</td></tr>
<tr><td rowspan="4">Liopholis inornata</td><td colspan="2">
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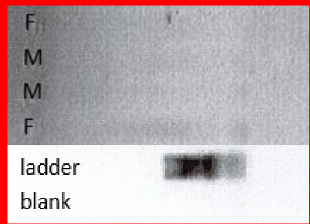
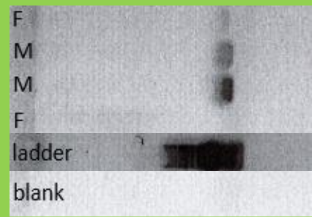
</td><td colspan="2">
</td><td colspan="2">
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<tr><td>55 – 60.5</td><td>NA</td><td>55.4 – 60.5</td><td>55.4 – 60.5</td><td>55 – 60.5</td><td>NA</td><td>58.5 – 60.5</td><td>59</td></tr>
<tr><td colspan="2">Notes: Band for only one male, bands for females. All bands <1000bp.</td><td colspan="2">Notes: Bands for males, no bands for females</td><td colspan="2">Notes: No bands</td><td colspan="2">Notes: Bands for males, putative band for one female</td></tr>
</table>

<table>
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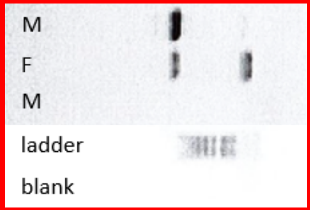
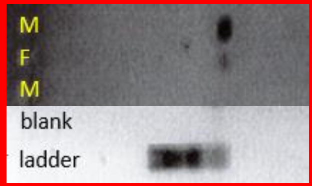
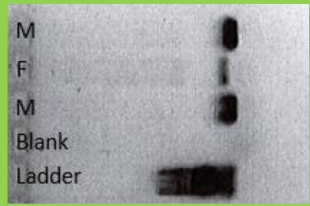
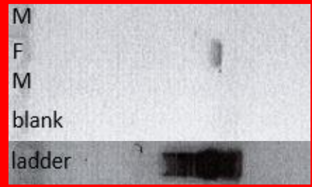
<td colspan="2"></td>
<td colspan="2"></td>
<td colspan="2"></td>
<td colspan="2"></td></tr>
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<tr><td>55 – 60.5</td><td>NA</td><td>55 – 60.5</td><td>NA</td><td>55 – 60.5</td><td>55 – 59.0</td><td>55 – 60.5</td><td>NA</td></tr>
<tr><td colspan="2">Notes: Bands for one male (>1000 bp) and the one female with an additional band for the female</td><td colspan="2">Notes: Band for one male, no band for the other male or females.</td><td colspan="2">Notes: Bands for males, putative band for female</td><td colspan="2">Notes: No band for males, putative band for the female</td></tr>
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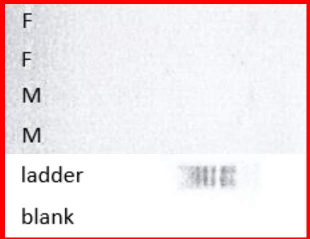
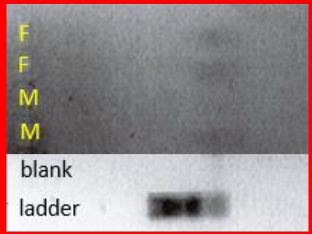
<td colspan="2"></td>
<td colspan="2"></td>
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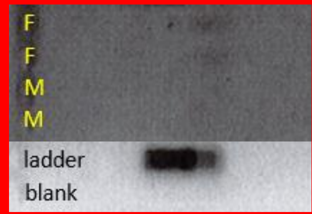
<td colspan="2"></td></tr>
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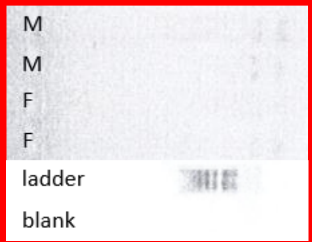
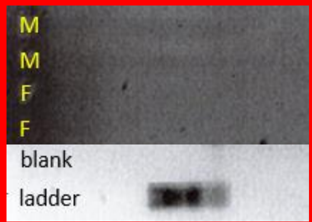
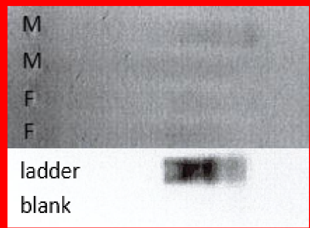
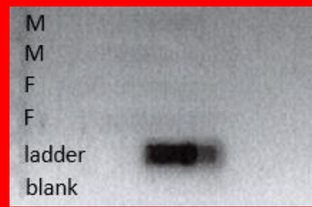
<td colspan="2"></td>
<td colspan="2"></td>
<td colspan="2"></td>
<td colspan="2"></td></tr>
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<table>
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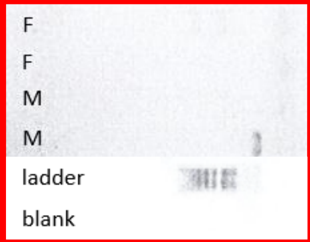
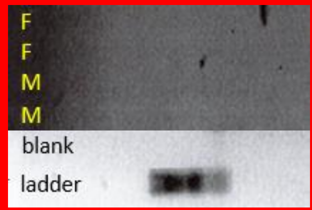
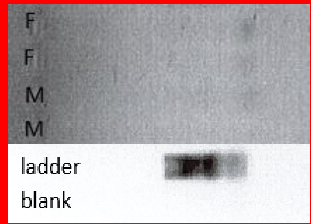
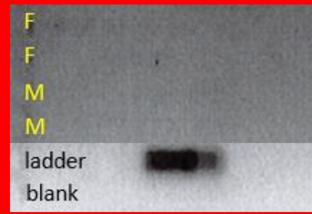
<tr><td rowspan="4">Carinascincus greenii</td><td colspan="2">
</td><td colspan="2">
</td><td colspan="2">
</td><td colspan="2">
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<tr><td colspan="2">Notes: No bands</td><td colspan="2">Notes: No bands</td><td colspan="2">Notes: No bands</td><td colspan="2">Notes: No bands</td></tr>
</table>

Figure S2: Gel images for the tests of the four primer pairs on the 13 species other than *L. whitii* within the Egernia group and the additional 3 species outside the Egernia group (subfamilies Mabuyinae and Eugongylinae). Sexes are indicated on each picture ('M' for males, 'F" for females). Each picture is associated to a ladder and a blank. All samples were run on a row including 23 other samples, a ladder and a blank. Background colour between some samples and their associated ladder differ because of different exposure across the gel. Green background indicates species/primer pairs that allowed for successful distinction between the sexes and therefore cross amplification. Red background indicates species/primer pairs where cross amplification failed.